

YINUO HAN

1200 E. California Blvd., Pasadena, CA 91125, USA
yinuo@caltech.edu · yinuohan.github.io

ACADEMIC APPOINTMENTS

1. **Barr Fellow, Caltech, USA** Since Oct 2024
Postdoctoral fellowship, Planetary Sciences, Division of Geological and Planetary Sciences

EDUCATION

1. **PhD in Astronomy, University of Cambridge, UK** Oct 2020 - Jul 2024
Institute of Astronomy · Trinity College · Gates Cambridge Scholar
Thesis: *The Structure of Extrasolar Planetesimal Belts in Images*
Advisor: Prof. Mark Wyatt
2. **BSc (Adv) (Hons), University of Sydney, Australia** Feb 2016 - Dec 2019
Majors in Physics, Neuroscience · First Class Honours (Physics) and the University Medal
Thesis: *A Dance with Dragons: The Enigmatic Wolf-Rayet Binary Apep*
Advisor: Prof. Peter Tuthill
3. **Study Abroad at University of California, Berkeley, USA** Aug 2017 - Dec 2017

RESEARCH INTERESTS

- Debris disks, planetary system formation and evolution
- Wolf-Rayet stars, colliding-wind binaries, carbon dust production
- High-resolution imaging, optical/infrared imaging, interferometry

ADVISING

List of theses and research projects advised.

- *VLT observations of β Pic and proper motion constraints of its dust clump.* Since Jun 2025
Summer research project (Caltech SURF) by Polaris Hayes.
- *ALMA imaging of WR112 and dust modeling with JWST/MIRI.* Since Jun 2025
Summer research project (Caltech SURF) by Donglin Wu.
- *Joint ALMA and JWST/NIRCam modeling of the vertical structure of the AU Mic debris disk.* Since Jun 2025
Summer research project (Caltech SURF) by Junyi Zhao.
- *Secular perturbation as the origin of the dust and gas clump of Beta Pictoris.* Oct 2022 – Jun 2023
Cambridge Part III/Masters project by Faran Sarwar (co-advised with Mark Wyatt).
- *The structure of the Beta Pictoris debris disk inferred from multi-wavelength mid-infrared images.* Jul – Aug 2022
Summer research project by Andrew Zhang.

TEACHING

1. **Topics in Astrophysics** (Distributions, Timescales, Tides, Planet Formation) 2023
Institute of Astronomy, University of Cambridge
Regular supervisions/tutorial classes for Part II Astrophysics/final-year undergraduate astrophysics.
2. **Structure and Evolution of Stars** 2022
Institute of Astronomy, University of Cambridge
Regular supervisions/tutorial classes for Part II Astrophysics/final-year undergraduate astrophysics.
3. **Oscillating Systems, Waves and Quantum Waves, Fields** 2021
Trinity College, University of Cambridge
Weekly supervisions/tutorial classes for Part IA Physics/first-year undergraduate physics.
4. **Physics 1B** (Electromagnetism, Fluids, Quantum Mechanics) 2019
School of Physics, University of Sydney
Weekly tutorial classes for first-year undergraduate physics.

AWARDS

UNIVERSITY OF CAMBRIDGE

- Gates Cambridge Scholarship* 2020
Cambridge International Scholarship, accepted Gates above instead 2020

UNIVERSITY OF SYDNEY

- University Medal* 2019
Henry Chamberlain Russell Prize for Astronomy for best astronomy Honours thesis 2019
University of Sydney Honours Scholarship 2019
Faculty of Science Olympiad Scholarship 2016 - 2019
Deas-Thomson Scholarship in Physics for top of class 2018
W.I.B. Smith Prize in Experimental Physics 2018
Dean's List of Academic Excellence 2016, 2017, 2018
Sydney Scholars Award 2016 - 2018
Denison Summer Research Scholarship 2016, 2017
Sydney Abroad International Exchange Scholarship 2017
University of Sydney Academic Merit Prize 2016
Levey Scholarship in Physics for top of class 2016
Smith Prize in Experimental Physics 2016

OTHER

- University of Oxford Clarendon Scholarship*, declined 2020
International Physics Olympiad Bronze Medal 2015

TALKS

RESEARCH TALKS

- Conference: Exploring Star–Planet–Disk Connections *Oct 2025*
High-resolution ALMA imaging of extrasolar planetesimal belts. *Tartu, Estonia*

- Seminar: MPIA Planet and Star Formation Seminar Oct 2025
The formation and evolution of dust in the colliding-wind binary Apep revealed by JWST. Heidelberg, Germany
- Seminar: Columbia Astronomy & Astrophysics Seminar Mar 2025
Resolving radial substructures in debris disks. New York City, USA
- Seminar: JPL Astrophysics Luncheon Seminar Feb 2025
Interpreting the Resolved Structure of Debris Disks. Pasadena, USA
- Conference: Europlanet Science Congress 2024 Sep 2024
The high-resolution structure of debris disks revealed by the ARKS ALMA program. Berlin, Germany
- Conference: Raising the veil on star formation near and far Apr 2024
High-resolution ALMA imaging of debris disks with the ARKS program. Cambridge, UK
- Conference: Dust Devils: Debris Disks in the Sonoran Desert Mar 2024
The radial structure of debris disks in the ARKS ALMA program. Tucson, USA
- Seminar: Department of Applied Mathematics and Theoretical Physics, University of Cambridge Oct 2023
The radial and vertical structure of debris disks. Cambridge, UK
- Conference: Cosmic Dust 2023 Aug 2023
The physics of dusty Wolf-Rayet binaries revealed by observations of WR140. Kitakyushu, Japan
- Seminar: Institute of Astronomy, University of Cambridge Oct 2022
Stellar fireworks: the spiral dust plumes of Wolf-Rayet binaries. Cambridge, UK
- Conference: UK Exoplanet Community Meeting 2022 Sep 2022
Recovering the structure of edge-on debris disks non-parametrically. Edinburgh, UK
- Conference: Debris Disks at Home and Abroad Sep 2022
Recovering the structure of edge-on debris disks non-parametrically. Jena, Germany
- Seminar: Institute of Astronomy, University of Cambridge Mar 2022
A non-parametric method to recover the structure of edge-on debris disks. Cambridge, UK

OUTREACH TALKS

- Institute of Astronomy Open Evening Oct 2023
What are exoplanets doing? Asteroid belts give us a clue. Cambridge, UK
- Institute of Astronomy Open Evening Jan 2023
Stellar fireworks: a journey through the spirally dust of massive binary stars. Cambridge, UK
- Integrated Science Class Seminar (Year 6–8), Ming Cheng School Apr 2022
Planets of the Solar System and beyond. Beijing, China

- Gates Cambridge Teach-a-thon Feb 2021
Planets of the Solar System and beyond. Online
- Australian Council of Undergraduate Research Oct 2019
Highly anisotropic stellar winds in recently-identified Wolf-Rayet binary. Newcastle, Australia

POSTER PRESENTATIONS

- Conference: Gordon Research Conference on the Origins of Solar Systems Jun 2025
The radial structure of debris disks in the ARKS ALMA program. South Hadley, USA
- Conference: Gordon Research Seminar + Conference on the Origins of Solar Systems Jun 2023
Modeling the structure of debris disks and the dust clump of Beta Pictoris. South Hadley, USA
- Conference: STScI Spring Symposium: Planetary Systems and the Origins of Life in the Era of JWST May 2023
Modeling the structure of debris disks and the dust clump of Beta Pictoris. Baltimore, USA
- Conference: Protostars and Planets VII Apr 2023
Has the dust clump in the debris disk of Beta Pictoris moved? Kyoto, Japan

OBSERVING

- *ALMA co-I. program:* Testing a new picture of debris disk gas by observing CI towards HD 121617 (ID 2025.1.00273.S, P.I. Gianni Cataldi) C.12, 2025
- *VLT/VISIR P.I. program:* Testing the giant impact scenario to explain the dust clump and cat's tail in Beta Pictoris (ID 114.27CQ) P114, 2024
- *ALMA P.I. program:* First resolved ALMA imaging of carbon-rich Wolf-Rayet binary dust structures (ID 2024.1.00803.S) C.11, 2024
- *ALMA co-I. program:* Resolving the extended structure associated with Apep's enigmatic wind (ID 2024.1.01244.S, P.I. Joseph Callingham) C.10, 2024
- *JWST P.I. program:* What lies beyond the inner spiral of Apep? (ID 5842) C.3, 2024
- *JWST P.I. program:* What causes warm dust interior to planetesimal belts? (ID 5709) C.3, 2024
- *VLT/VISIR P.I. program:* Wind kinematics in the extreme colliding-wind binary Apep: the first Galactic long-duration gamma-ray burst progenitor? (ID 113.26A7) P113, 2024
- *ALMA P.I. program:* First ALMA imaging of dust around Wolf-Rayet binaries (ID 2023.1.00999.S) C.10, 2023
- *ALMA co-I. program:* Vertical structure and planetary system dynamics (ID 2023.1.00487.S and 2025.1.00362.S, P.I. Meredith Hughes) C.10/12, 2023/2025
- *JWST co-I. program:* Fingerprinting the history of episodic dust creation in Wolf-Rayet binaries (ID 4093, P.I. Noel Richardson) C.2, 2023
- *Australia Telescope Compact Array:* Executing observations of Wolf-Rayet binary Apep 2019

ADDITIONAL EXPERIENCE

REFEREE & PEER REVIEW

Referee for MNRAS and AAS Journals
ALMA distributed peer review, Cycles 8 – 12
ESO distributed peer review

GRANTS

P.I., JWST Program 5842 (USD 78k) 2024
P.I., JWST Program 5709 (USD 74k) 2024

COLLABORATIONS

ARKS: ALMA Large Program to image debris disks at high resolution, P.I. Sebastian Marino
• *Leading Radial Structure team*
JWST High Contrast Imaging: JWST ERS program imaging exoplanets and disks, P.I. Sasha Hinkley
• *Nonparametric modeling of JWST 4-quadrant phase mask observations and application to HD 141569*
DustERS: JWST ERS program on Wolf-Rayet binary dust formation, P.I. Ryan Lau
• *Column density modeling of MIRI imaging and leading follow-up proposals*

PRESS

NASA, STScI and Caltech press releases based on Han et al. 2025. Nov 2025
Featured on Astronomy Picture of the Day.
University of Cambridge press release based on Han et al. 2022. Oct 2022
Reported by major science and mainstream news outlets (e.g., ABC, BBC, CNN, The Guardian).
University of Sydney press release based on Han et al. 2020. Oct 2020
Reported by 110 news items from 21 countries in 8 languages (e.g., CNN).

OTHER EMPLOYMENT

Supervisor (tutorial teaching), Institute of Astronomy, Cambridge 2022 - 2023
Supervisor (tutorial teaching), Trinity College, Cambridge 2020 - 2021
Tutor (tutorial teaching), School of Physics, University of Sydney 2019
Strategist, Goldman Sachs 2018

COMMUNITY INVOLVEMENT & OUTREACH

Public talk speaker, Institute of Astronomy, Cambridge open evenings 2022 - 2023
Technology Officer, Gates Cambridge Scholars Council 2020 - 2021
General Executive, Sydney University Mathematics Society 2017 - 2019
Ambassador, Australian Science Innovations 2016 - 2019
Interviewer, Berkeley Scientific Journal 2017

PUBLICATIONS

FIRST-AUTHOR PUBLICATIONS

7. [In press] **Y. Han**, E. Mansell, J. Jennings, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) II. The Radial Structure of Debris Disks. *Astronomy & Astrophysics* (2025).

6. **Y. Han**, R. M. T. White, J. R. Callingham, *et al.* The formation and evolution of dust in the colliding-wind Wolf-Rayet binary Apep observed by JWST. *The Astrophysical Journal* (2025). doi.org/10.3847/1538-4357/ae12e5
5. **Y. Han**, M. C. Wyatt, S. Marino. Recovering the structure of debris disks non-parametrically from images. *Monthly Notices of the Royal Astronomical Society* (2025). doi.org/10.1093/mnras/staf282
4. **Y. Han**, M. C. Wyatt, W. R. F. Dent. Has the dust clump in the debris disk of Beta Pictoris moved? *Monthly Notices of the Royal Astronomical Society* (2023). doi.org/10.1093/mnras/stac3769
3. **Y. Han**, P. G. Tuthill, R. M. Lau, A. Soulain. Radiation-driven acceleration in the expanding WR140 dust shell. *Nature* (2022). doi.org/10.1038/s41586-022-05155-5
2. **Y. Han**, M. C. Wyatt, L. Matrà. RAVE: a non-parametric method for recovering the surface brightness and height profiles of edge-on debris disks. *Monthly Notices of the Royal Astronomical Society* (2022). doi.org/10.1093/mnras/stac373
1. **Y. Han**, P. G. Tuthill, A. Soulain, J. R. Callingham, P. M. Williams, P. A. Crowther, B. J. S. Pope, B. Marcote. The extreme colliding-wind system Apep: resolved imagery of the central binary and dust plume in the infrared. *Monthly Notices of the Royal Astronomical Society* (2020). doi.org/10.1093/mnras/staa2349

IN PREPARATION AND IN REVIEW

2. [*In prep.*] **Y. Han**, M. C. Wyatt, K. Y. L. Su, *et al.* A radially broad collisional cascade in the debris disk of γ Oph revealed by JWST.
1. [*In prep.*] **Y. Han**, M. C. Wyatt, M. Jankovic, *et al.* The vertical structure of the Beta Pictoris debris disk viewed at mid-infrared and mm wavelengths.

CO-AUTHORED PUBLICATIONS

20. [*In press*] S. Marino, L. Matrà, A. M. Hughes, J. Ehrhardt, G. M. Kennedy, C. del Burgo, A. Brennan, **Y. Han**, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS). I: Motivation, Sample, Data Reduction and Results Overview. *Astronomy & Astrophysics* (2025).
19. [*In press*] B. Zawadzki, A. Fehr, A. M. Hughes, E. Mansell, J. Kittling, **Y. Han**, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS). III: The Vertical Structure of Debris Discs. *Astronomy & Astrophysics* (2025).
18. [*In press*] S. Mac Manamon, L. Matrà, S. Marino, A. Brennan, **Y. Han**, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS). IV: CO gas imaging and overview. *Astronomy & Astrophysics* (2025).
17. [*In press*] J. B. Lovell, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS). VI: Asymmetries and Offsets. *Astronomy & Astrophysics* (2025).
16. [*In press*] P. Weber, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS). IX: Gas-driven origin for the continuum arc in the debris disc of HD121617. *Astronomy & Astrophysics* (2025).

15. [In press] M. Jankovic, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS). X: Interpreting the peculiar dust rings around HD 131835. *Astronomy & Astrophysics* (2025).
14. R. M. T. White, B. J. S. Pope, P. G. Tuthill, **Y. Han**, *et al.* The Serpent Eating Its Own Tail: Dust Destruction in the Apep Colliding-Wind Nebula. *The Astrophysical Journal* (2025). doi.org/10.3847/1538-4357/adfbel
13. J. D. Monnier, **Y. Han**, *et al.* Revealing the Accelerating Wind in the Inner Region of Colliding-wind Binary WR 112. *The Astrophysical Journal* (2025). doi.org/10.3847/1538-3881/adfa03
12. N. D. Richardson, *et al.* Carbon-rich dust injected into the interstellar medium by Galactic WC binaries survives for hundreds of years. *The Astrophysical Journal* (2025). 10.3847/1538-4357/addf30
11. M. Sommer, M. C. Wyatt, **Y. Han**. A PR drag origin for the Fomalhaut disk’s pervasive inner dust: constraints on collisional strengths, icy composition, and embedded planets. *Monthly Notices of the Royal Astronomical Society* (2025). doi.org/10.1093/mnras/staf494
10. E. P. Lieb, *et al.* Dynamic imprints of colliding-wind dust formation from WR140. *The Astrophysical Journal Letters* (2025). doi.org/10.3847/2041-8213/ad9aa9
9. R. M. Lau, *et al.* A first look with JWST aperture masking interferometry (AMI): resolving circumstellar dust around the Wolf-Rayet binary WR 137 beyond the Rayleigh limit. *The Astrophysical Journal* (2024). doi.org/10.3847/2041-8213/ad9aa9
8. J. Terrill, S. Marino, R. A. Booth, **Y. Han**, J. Jennings, M. C. Wyatt. Deprojecting and constraining the vertical thickness of exoKuiper belts. *Monthly Notices of the Royal Astronomical Society* (2023). doi.org/10.1093/mnras/stad1847
7. R. M. Lau, *et al.* From dust to nanodust: resolving circumstellar dust from the colliding-wind binary Wolf-Rayet 140. *The Astrophysical Journal* (2022). doi.org/10.3847/1538-4357/acd4c5
6. R. M. Lau, M. J. Hankins, **Y. Han**, *et al.* Nested dust shells around the Wolf-Rayet binary WR 140 observed with JWST. *Nature Astronomy* (2022). doi.org/10.1038/s41550-022-01812-x
5. P. A. Robinson, X. Gao, **Y. Han**. Relationships between lognormal distributions of neural properties, activity, criticality, and connectivity. *Biological Cybernetics* (2021). doi.org/10.1007/s00422-021-00871-z
4. B. Marcote, J. R. Callingham, M. De Becker, P. G. Edwards, **Y. Han**, R. Schulz, J. Stevens, P. G. Tuthill. AU-scale radio imaging of the wind collision region in the brightest and most luminous non-thermal colliding wind binary Apep. *Monthly Notices of the Royal Astronomical Society* (2021). doi.org/10.1093/mnras/staa3863
3. R. M. Lau, M. J. Hankins, **Y. Han** *et al.* Resolving periodic spirals and shadows from the Wolf-Rayet dust factory WR112. *The Astrophysical Journal* (2020). doi.org/10.3847/1538-4357/abaab8
2. J. R. Callingham, P. A. Crowther, P. M. Williams, P. G. Tuthill, **Y. Han**, B. J. S. Pope. Two Wolf-Rayet stars at the heart of colliding-wind binary Apep. *Monthly Notices of the Royal Astronomical Society* (2020). doi.org/10.1093/mnras/staa1244

1. M. Li, **Y. Han**, M. J. Aburn, M. Breakspear, R. Q. Poldrack, J. M. Shine, J. T. Lizier. Transitions in information processing dynamics at the whole-brain network level are driven by alterations in neural gain. *PLOS Computational Biology* (2019). doi.org/10.1371/journal.pcbi.1006957